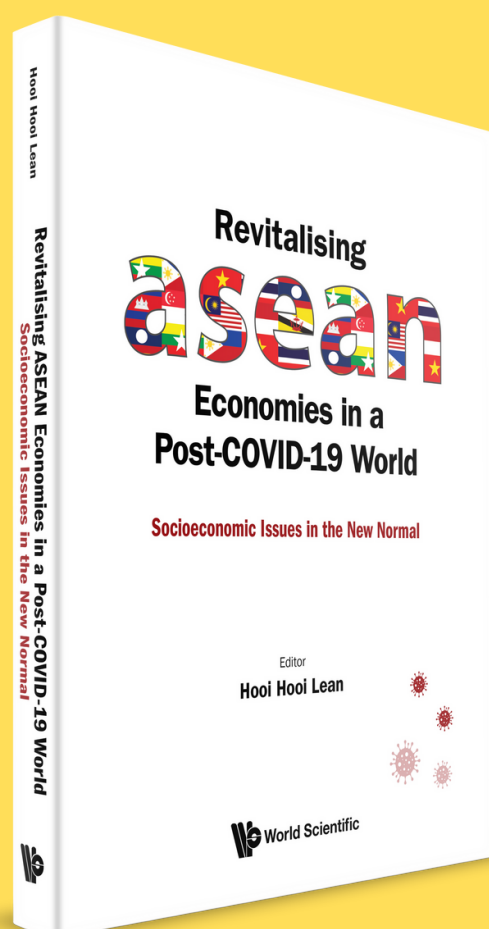


REVITALISING asean ECONOMIES IN A POST-COVID-19 WORLD

Socioeconomic Issues in the New Normal



edited by
Hooi Hooi Lean
Universiti Sains Malaysia, Malaysia



284pp | June 2022

Hardcover 978-981-122-846-9 | **US\$98 / £80 / S\$145**

eBook-Individuals 978-981-122-848-3 | **US\$78 / £60 / S\$116**

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The novel coronavirus (COVID-19) pandemic was reported in China in December 2019, and later spread to other parts of the world. Countries in Southeast Asia were some of the first nations affected by the pandemic due to their geographical proximities and trade relations with China. In January 2020, the virus spread to some countries within the Association of Southeast Asian Nations (ASEAN). The number of confirmed cases kept increasing within ASEAN and other countries of the world. Consequently, countries in ASEAN introduced preventive and containment measures to cope with the pandemic, which include quarantines, lockdowns, restrictions of movements and large gatherings, as well as school and business closures.

Beyond the efforts of preventing and containing the spread of the virus, ASEAN's most significant challenges are the social and economic crisis of historic proportions that are beginning to unfold. COVID-19 pandemic has brought interruptions in all sectors of ASEAN economies. It is, therefore, imperative to assess the extent to which the pandemic has impacted the social and economic aspects of the region. This book intends to highlight the social and economic impacts of the COVID-19 pandemic in ASEAN and suggest ways on how to mitigate them. It covers how the shocks from COVID-19 have impacted production, supply chains, demand, commodity prices, consumer behaviour, financial markets, employment, services, transportation and community, and the various policies that ASEAN authorities should put in place to mitigate the effects of the pandemic.

Readership

Academic, researchers, postgraduate students, undergraduate students, industry and government agencies interested in Asian economies and its development and international relations.

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About the Editor

Hooi Hooi Lean is a professor of Economics at the School of Social Sciences, Universiti Sains Malaysia. She obtained her honors bachelor degree of Economics majoring in Statistics from the University of Malaya, Master of Science (Statistics) from the Universiti Sains Malaysia, and Ph.D. (Economics) from the National University of Singapore. She served as a post-doctoral visiting scholar to the Department of Economics at Monash University, Australia. Dr. Lean has been a visiting scholar at a number of prestigious international universities, including Sungkyunkwan University (Korea), Tamkang University (Taiwan), CEREFIGE Université Nancy 2 (France), Auckland University of Technology (New Zealand), Georg-Simon-Ohm University of Applied Sciences (Germany), and University of Rennes 2 (France).

Prof. Lean has authored more than 170 scholarly articles and academic book chapters. Her works have been published in many reputed international journals and publications. Her H-index is 46 and there are more than 9,000 citations to her works on Google Scholar. She has also presented on various topics at many major international conferences held worldwide. In addition to these talks in the academic circuit, she has been invited to give talks at some private corporations, organisations, and educational institutions. She has also provided consultancy works to the private sector, government agencies, and non-governmental organisations.

Prof. Lean has been recognised as among the Top Research Scientists Malaysia by Academy of Sciences Malaysia in 2018. She received

Malaysia's Research Star Award in 2017 from the Ministry of Higher Education Malaysia and Clarivate Analytics, and the National Academic Award in 2015 from the Ministry of Higher Education Malaysia. She was listed as top 2% scientists in the world by Stanford University. She has been listed in the Who's Who in the World since 2009 and granted the prestigious title of "Researcher of the Week" in GDN East Asia in 2012 for her excellent contributions. She has been awarded the ASEAN-ROK Academic Exchange Fellowship Programme in 2007, the Democratic Pacific Union (DPU) Visiting Fellowship in 2008, the International HERMES Fellowship Programme in 2009, and Faculty Exchange Fellowship from Georg-Simon-Ohm University in 2012. Dr. Lean has also won the "Sanggar Sanjung" Excellent Award for Publication continuously since 2009, the "Hadiah Sanjungan" Best Award for Publication 2006–2011, and the Excellent Service Award 2010 & 2014 from the Universiti Sains Malaysia. Dr. Lean has received Newton Research Grant from Research Councils United Kingdom; the Fundamental Research Grant Scheme and Research University Grant from the Ministry of Higher Education Malaysia.

Prof. Lean is a fellow of the East Asian Economic Association. She serves as an associate editor for the *Singapore Economic Review*, *Frontiers in Energy Research*, *Malaysian Journal of Economics*, and the *International Journal of Economics and Management*. She is also an editorial board member of *Asian Journal of Economics and Finance*, *Energy Research Letters*, *Journal of Asian Finance, Economics and Business*, and *Journal of Risk and Financial Management*. She is also on the editorial board as an advisory member of the *Indonesian Capital Market Review* and the *Labuan Bulletin of International Business & Finance*. She has reviewed more than 200 papers for various academic journals and received Peer Review Award from Publons in 2018. She has supervised many undergraduate and postgraduate students and has organised several international conferences/seminars/workshops. Besides her work in the academic sphere, she also contributes towards the various multi-cultural communities in Malaysia and religious development. She holds several private or community-based positions such as Deputy President of Malaysia Silk Road Society, Consultative Panel of Young Buddhist Association of Malaysia Penang State Liaison Committee, Board of Governor of the Penang Buddhist Association Dharma Sunday School, a committee member of the Penang Buddhist Association.

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https://doi.org/10.1142/9789811228476_0007

Chapter 7

Impacts of COVID-19 Pandemic on Thai Fruit: A Case Study of Longan Supply Chain

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Abstract

COVID-19 has had tremendous ripple effect upon every sector in the system since its beginning in 2019. Not only the effect of the pandemic itself, but the preventive measures of the government also affected the system, especially the agriculture supply chain widely considered to be the most important part of Thailand. The aims of the research chapter were to investigate and analyse the effects of COVID-19 and government measures towards the fruit supply chain in Thailand, together with reviewing on-season longan as a case study. Consequently, the instrument utilised was the supply chain mapping. The result was collected additionally by in-depth interviews, focus-group interviews, and questionnaires from Thai farmers. As a result, COVID-19 pandemic and preventive measures indicated the entire fruit supply-chain system's impacts on activities and stakeholders from start to finish nationally and internationally. Besides, the labour markets were affected, particularly seasonal foreign harvesting workers. These effects are in accordance with the agricultural overproduction, farm price and export price decreases, and other export volumes. In case of on-season longan, the shortage of fruit harvesting workers from disease preventive measures, including the prohibition of cross-border travel, were highlighted as significant problems. Additionally, the 14-day quarantine measure lessened the number of brokers, traders, and agricultural workers. These problems put pressure on involved people and changed the consumer behaviour. In sum, the noteworthy results lead to the policy recommendations for the new normal of the marketplace, comprising promoting comprehensive learning and training (reskill, upskill, and new skill) courses for agricultural practices and so forth.

Keywords: COVID-19; pandemic; agriculture; supply chain; Thailand; fruit; longan.

1. Introduction

The ongoing situation of the coronavirus disease 2019 (COVID-19) pandemic known as the coronavirus pandemic was first identified and declared in December 2019. Later, the World Health Organization (WHO) declared the outbreak a Public Health Emergency of International Concern

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in January 2020 and announced it a pandemic in March 2020. The governments in several countries have immediately responded to COVID-19 with a number of different non-pharmaceutical measures to prevent infections and the spread of COVID-19. Most of the preventative measures are curfew declarations or lockdowns in varying degrees (i.e. throughout the country or particular region, province, district, or community), prohibiting mass public gatherings, closing public spaces, restricting public transportation, implementing Working from Home (WFH) arrangements, as well as the border closure together with suspension of commercial international flights. Despite passing the legislation and measures in response to the pandemic, these measures disrupt the global supply chain and logistic services of agriculture and food systems which is the crucial key and needed most in order to access the essential agriculture and food resources as well as other key necessities.

Due to the economic consequences of the ongoing COVID-19 pandemic, economic depression began in numerous nations (Jackson *et al.*, 2020; Kanu, 2020). The COVID-19 recession appears to be severe and has become the worst global economic crisis since the Great Depression of the 1930s. Thus, it is of great significance that COVID-19 inevitably has impacted the global agriculture and food supply chain, which is one of the most important sectors of the economy (IMF, 2020; Ivanov, 2020). COVID-19 resulted in the movement restrictions of agricultural labour supply and transportation, changes in consumer demand and purchasing behaviour for food products, shutdowns of food manufacturers and service providers, shipping disruption and temporary port closures, increased costs from public health measures requirements, and financial pressures in food supply chain (Aday and Aday, 2020). All of these situations lead to the intensive increase in global food insecurity, economic and social impacts, which affect the well-being of households presently and almost every country subsequently.

Among initial measures, many governments are concerned about guaranteeing food availability for domestic consumption, food security, and alleviating problems for agricultural and food manufacturers to survive under many restrictions. Owing to the significance of food for life and income-inelasticity of food demand, trade in agricultural products has

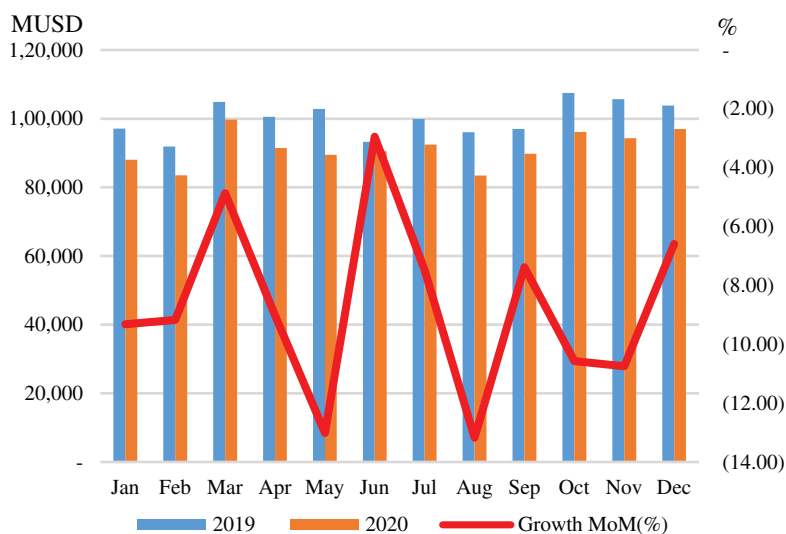
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Figure 1: The value and growth of global agriculture export of 2019 and 2020.

Source: Trade Map (2020).

been more resilient than other sectors in the system, such as industry and tourism sectors.

Figure 1 presents the value and growth rate of total global agriculture export of 2019 and 2020. In 2020, the total value of global agricultural export (HS 01-21) decreased to USD 1,095.8 million compared with USD 1,200.4 million from 2019, or equivalent to a diminishing of -8.72%. The export value growth rates have shown negative signs since the beginning of the COVID-19 pandemic. In January 2020, the agriculture export value shrank from USD 97.1 million to USD 88 million or decreased 9.32% compared with the same month in 2019. On account of the ongoing COVID-19 pandemic, the inability to control the pandemic in some regions with long-lasting economic impacts affects the consumer confidence index. The line represents the global agriculture export value growth rate in comparison with the same period between 2019 and 2020. The result shows that all have negative values, especially in August, which has the highest negative growth rate at -13.17%.

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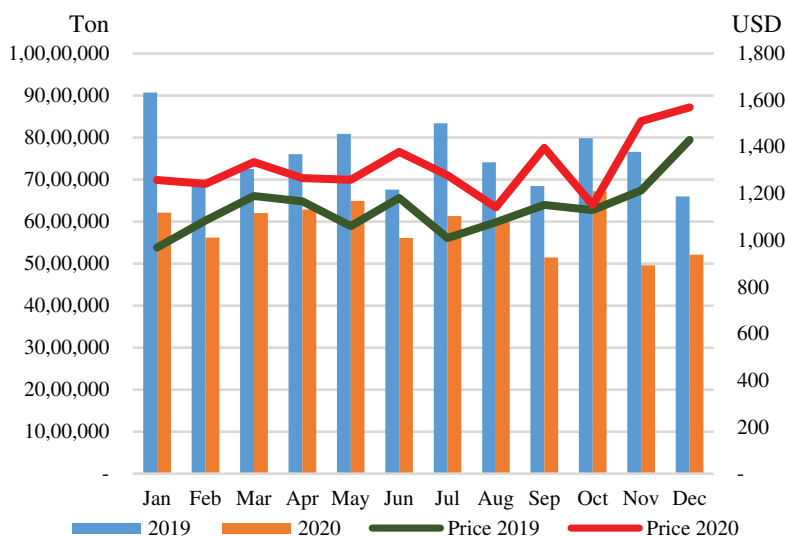


Figure 2: The volume and price of global edible fruit export of 2019 and 2020.

Source: Trade Map (2020).

The total volumes and average prices of global edible fruit export are displayed in Figure 2. The overall global edible fruit export volume in 2020 was less than in 2019, while its average price increased. The highest percentage average price change in the same period between 2019 and 2020 was 29.8% in January followed by 26.55% and 24.5% in July and November, respectively. The main causes are simultaneous disruptions in supply, demand, supply chain, and logistics infrastructure due to the COVID-19 pandemic. In addition, edible fruits normally have a greater price and income elasticity than staple food that provides more energy and calories than fruits, such as grains, rice, wheat, corn, and cassava (Andreyeva *et al.*, 2010; Femenia, 2019).

It is noted that the decrease in agricultural export has arisen from several causes. All of them relate to the many activities and stakeholders in the agricultural supply chain. To illustrate, labour sickness and shortage, and social distancing measures in some countries significantly reduced agricultural labour supply leading to the loss of agricultural production and food security (World Bank, 2020). They accordingly forbid

food export in order to ensure sufficient food for domestic consumption (Aday and Aday, 2020). For example, the governments of Vietnam and Cambodia were considerably concerned about food security in their countries, therefore they placed an export ban on their rice. In addition, cross-border shipping delays, interruptions in logistics, and restrictions on exports from various measures in each country cause a decline in export volumes. The increase in export cost resulted in higher export prices correspondingly (Evenett *et al.*, 2020). Most hotels, restaurants, catering, and cafes were closed as a result of the government measures in provision for demand reductions of fruits and vegetables. Although food demand, especially of fresh fruits and vegetables, has lessened; food-security concerns to food importing countries, in terms of the lower production volumes and restrictions on exports, pushed the global food price up by 18% on average (Espitia *et al.*, 2020).

Moving to the case of Thailand, the first case of COVID-19 was confirmed on 13 January 2020. The Prime Minister declared a state of emergency, effective on 26 March 2020, and a curfew went into effect on 3 April 2020. All commercial international flights were suspended from 4 April 2020; consequently, lockdown measures were implemented in varying degrees throughout the country. Measures as mentioned have had a direct impact on plenty of economic activities, including production of goods and services, household consumption, private businesses and commerce, along with transportation and logistics.

Thai tropical fruits are considered an important economic crop in Thailand, generating billions of US dollars each year (more than USD 3.6 billion in 2019) and popularly consumed both domestically and internationally. It is noteworthy to mention that Thai tropical fruits are economically important and have high export value, specially longans, durians, mangosteens, mangoes, rambutans, etc. Figure 3 shows the monthly value and growth of Thailand edible fruit export. In the first 2 months of 2020, it was pointed that the export values decreased, and the export value growth rates were negative. However, between March to May, which is the prior and high season for tropical fruits of Thailand, the export value increased from THB 327 million in March to THB 785 million in May. The export value growth rates of March and May compared to the same period between 2019 and 2020 were 13.68% and 143.82%, respectively.

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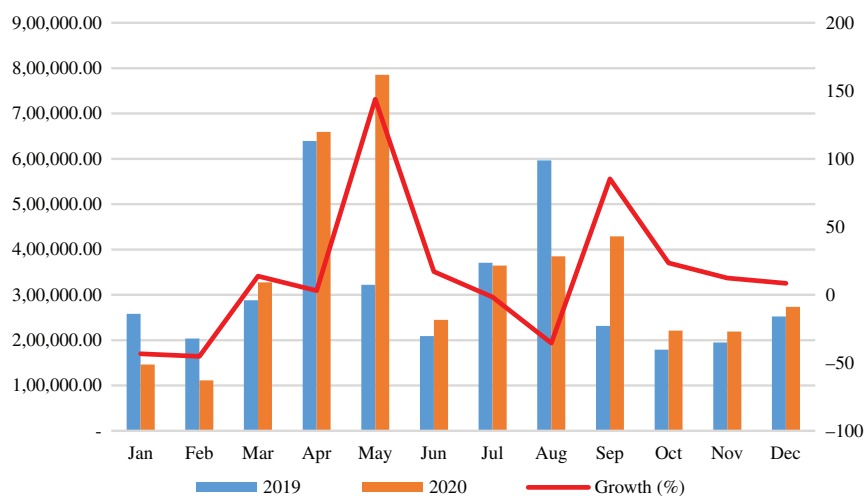


Figure 3: Value and growth of Thai edible fruit export.

Source: Trade Map (2020).

To shed light on this issue, this chapter firstly aims to document and analyse what the impacts of COVID-19 are on Thai fruits, and in what way the government can alleviate the fruit growers and stakeholders. According to this study, most Thai fruits were adversely affected by the COVID measures, excluding durians. It should be noted that the major problems in Thai fruit supply chain are labour shortages, lower farm-gate prices, increased domestic and international transport restrictions, including sanitary requirements, rising freight costs, and lower consumer purchasing power (Attavanich, 2020; Suriyankietkaew *et al.*, 2020).

Besides, longan is a tropical fruit that is classified as the main economic fruit and product champion of Thailand that can be produced for domestic and international consumption, with the second largest export value after durian, holding China as the main export market. While durian export values increased compared with the previous year, longan export values dropped significantly. Therefore, the research question is what the economic and social impacts of COVID-19 are on in-season longan supply chain in Thailand.

In short, the specific objectives are: (1) to systematically review and empirically investigate the effects of COVID-19 and the government

alleviation strategies regarding it on the fruit sectors in Thailand considering different aspects, (2) to study the impacts of COVID-19 pandemic on in-season longan supply chain in Thailand, and (3) to provide the possible balance recovery policies in order to alleviate the impact of COVID-19 on stakeholders in Thai fruit supply chain. This chapter is organised as follows. Section 1 gives a brief overview of the rationale of the chapter and objectives. Section 2 is literature about the impact of previous pandemics and the COVID-19 pandemic on agricultural sector and supply chain, emphasising on fruits. In Section 3, the research methodology and data collection are presented. Section 4 shows the empirical results. Finally, the conclusion and policy recommendations are stated in Section 5.

2. Literature Review

In the last 30 years prior to the COVID-19 pandemic, there have been global outbreaks affecting people, the agricultural sector, and food supply chain, such as H5N1, H7N9, H1N1, SARS, MERS, and Ebola. The first group is a group of influenza A viruses (H5N1, H7N9, and H1N1) which directly affect the meat production used as human food source, and meat supply chain. The avian influenza virus (H5N1 and H7N9), also known as avian flu or bird flu, is a concern due to the global spread of H5N1 that constitutes a pandemic threat. It can be transmitted between different birds and poultry, and to humans. The H5N1 and H7N9 outbreaks have had a negative impact on the poultry meat industry and supply chain, and poultry trade in many countries, particularly major export countries such as China, USA, and Thailand (Taha, 2007; Nicita, 2008; Mohan *et al.*, 2009; Kumar and Chandra, 2010; Kumar, 2012; Khokhar *et al.*, 2015). Shifting consumer food choices have decreased poultry consumption, resulting in lower domestic poultry prices (Taha, 2007; Beach, 2008). H5N1 was first confirmed in poultry and humans in Thailand in January 2004 (Tiensin *et al.*, 2005). Between 2004–2006, the H5N1 caused 25 human cases and 17 deaths in Thailand. Thailand had a 75% drop in poultry exports in the first quarter of 2004 (Blayney *et al.*, 2006; Taha, 2007).

In 2009, H1N1 influenza pandemic, also known as swine flu pandemic, was found in Mexico, and later was confirmed as human cases of infection spread throughout Mexico, in parts of the United States, and in

several countries worldwide. It can be transmitted between swine, and to humans. H1N1 also affected the pig industry, pork supply chain, and exports (Rassy and Smith, 2013). At the initial stage, it immediately caused a downturn in domestic and international pork markets. Domestic pork demand and prices dropped sharply because of food safety concerns of consumers. Several pork-importing countries such as China, Indonesia, and Azerbaijan, banned and restricted pork imports from certain countries. H1N1 pandemic effect also spilt over to other agricultural markets in the supply chain, such as feed grain and other livestock markets.

The second group is a group of coronaviruses which are respiratory infectious diseases such as severe acute respiratory syndrome coronavirus (SARS-CoV or SARS-CoV-1) in 2003, Middle East respiratory syndrome-related coronavirus (MERS-CoV) in 2009, and coronavirus disease 2019 (COVID-19 or SARS-CoV-2). In many countries, the situation worsens as the disease progresses. Therefore, many measures such as lockdowns, social distancing, movement restrictions, public health measures, and social and economic measures were implemented by governments worldwide in response to the pandemic. These measures indirectly affected agriculture and food supply chain through restricted tourism, transportation, and labour movement in the agricultural sector. In case of the MERS outbreak in the Republic of Korea, measures indirectly affected the agriculture and food supply chain through tourism, and travel-related service sectors, including food and beverages, and transportation (Joo *et al.*, 2019). SARS also has a significant impact on domestic consumption, tourism, and related industries, as is the spread of SARS in China and other countries. For instance, it intervenes in the household food demand and consequently causes the drastic decline in the number of hotel guests (Siu and Wong, 2004; Qiu *et al.*, 2018).

In case of Ebola virus disease, De La Fuente (2020) studied the impact of the West African Ebola epidemic on agricultural production and rural welfare, and found that the 2014–2015 Ebola epidemic had devastated both humans and economies. The agricultural sector of three countries, namely Sierra Leone, Guinea, and Liberia, has been severely affected by the Ebola pandemic (Qureshi, 2016). The higher Ebola prevalence disrupted the mobilisation of labour for cultivation and harvest, and many farms were abandoned. This resulted in a decrease in rice cultivation

areas and rice yields that may in turn have led to food price increases. Household welfare, as measured by per capita expenditures, was much lower in Ebola-endemic areas, especially those areas that were negatively impacted by a shortage of agricultural workers.

In case of COVID-19, there are considerable research papers reviewing and anticipating the impacts of COVID-19 on the agriculture sector. It impacts agriculture in two key areas: demand and supply for food (Siche, 2020). In accordance with Pan *et al.* (2020), studying the influence of COVID-19 on agricultural economy and emergency relief measures in China, plenty of impacts on many aspects were highlighted such as production, product supply, income and employment, and product sales model and trade. In addition, COVID-19 pandemic control measures have delayed and interrupted transport and logistics services. The border closures and further procedures, together with investigations, create congestion and delays affecting the transport of perishable products (OECD, 2020).

The declaration of export restrictions and more sanitary requirements in several countries restricted the global agri-food trade and market access. The COVID-19 outbreak has had a broader impact, bringing about a continued decline in agricultural products (Espitia *et al.*, 2020; Miller, 2020; Pulubuhu *et al.*, 2020). To illustrate, food products seem to be expensive due to transport restrictions and the steadily declining volumes of agricultural products (Pulubuhu *et al.*, 2020). Additionally, the COVID-19 epidemic could affect the availability of important intermediate inputs for farmers. Low inputs and/or high prices, such as pesticides, could negatively affect crop yields in 2020 and 2021, especially in developing countries (Schmidhuber *et al.*, 2020).

The number of labour in the agricultural sector tends to decline in terms of availability, number of labourers, and labour productivity (Gong, 2020; World Bank, 2020) due to labour sickness and shortage, 14-day quarantine measures, and restrictions on the movement of workers across the country and specific areas. Government guidance of urging residents not to leave home has incommoded plenty of farmers from doing their activities, leading to lower incomes (Pulubuhu, 2020). Hence, restrictions on the movement of people across borders have contributed to labour shortages in agriculture in many countries, particularly during peak seasonal labour demand or labour-intensive production (OECD, 2020).

In case of Thailand, Attavanich (2020) has evaluated and reported the impacts of COVID-19 on food systems, food security and nutrition, and livelihoods. From in-depth interviews, focus group and survey data from 1,449 farm households and 1,345 general households across the country showed that farm households likely received higher negative impact than general households. The COVID-19 preventive and containment measures of Thai government caused migrant labour shortage of fruit harvesting, food supply chain disruption, contraction in domestic consumption, the delay of exports or inability to export to international market from Thai-border trade checkpoints closing and raising of transportation, logistic, and shipping cost, which lead to an oversupply problem. The fresh fruit farmers were the most affected compared to other stakeholders in the fruit supply chain system. The report revealed that the highest income per capita reduction among Thai tropical fruits is longan farm households.

Concerning perspectives of ASEAN community, many countries in ASEAN depending on agriculture also responded immediately to COVID-19 pandemic. Suriyankietkaew *et al.* (2020) reported that each ASEAN country has different measures to deal with this pandemic. Many key sectors such as agriculture, supply chain and logistic, and tourism have been affected by lockdown and other measures. The lockdown damaged not only the incomes, employment, and social welfare but also food security and sustainability of the country. Although ASEAN response was slow and lacked consistency in the early stage, ASEAN later proposed the Declaration of the Special ASEAN Summit on Coronavirus Disease 2019 which has some measures related to recuperate the agricultural sector, to maintain supply chain connectivity, and to take collective actions and coordinate policies in alleviating the economic and social impact, safeguarding the people's well-being and preserving social economic stability in the face of the pandemic.

3. Research Methodology and Data Collection

This study will be conducted to gather relevant literature, documents, and information about impact of COVID-19 crisis on Thai fruit sector in macro views, and primary data from multi-lateral stakeholders in the Thai fruit supply chain. Data will be derived from various data collection

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methods, including (1) primary data from in-depth interviews and focus group interviews from multiple stakeholders and questionnaires, (2) secondary data from recent reports and related statistics from government agencies (The Office of Agricultural Economic, National Statistical Office, Ministry of Agriculture and Cooperatives, and Ministry of Commerce), and private agencies (CEIC database, Trade Map database) plus research reports from academic institutions, and international organisations.

Primary data are used in order to gain insightful perspectives and better understanding about the situations and contexts. This research proceeded with in-depth interview with key opinion leaders and influential stakeholders with various parties of fruit and longan supply chains in seven provinces, comprising Chiang Mai, Lamphun, Loei, Samut Sakhon, Chanthaburi, Sa Kaeo, and Songkhla. The stakeholders are fruit growers, local collectors, contract merchants, cooperatives, middle men (Lhong), wholesalers, retailers, and exporters totalling 58 observations. In terms of longan, this study provides focus group interviews with several stakeholders in the longan supply chain in Chiang Mai and Lamphun province (40 participants) in order to observe their opinions on the issues and impacts of the coronavirus situation on in-season longan supply chain. It is remarkable that Chiang Mai and Lamphun provinces have the most in-season longan cultivation in Thailand and represent the in-season longan supply chain in the northern part of Thailand.

In addition, 200 questionnaires were collected from longan growers in Chiang Mai and Lamphun provinces by random sampling method, where the major in-season longan plantations are, in order to investigate the impacts of COVID-19 pandemic on in-season longan farmers. Inexplicably, it is crucial to note that only in Chiang Mai and Lamphun provinces the productive area of longan is 92,531 hectares, accounting for almost 50% of the total areas (Office of Agricultural Economics, 2020a). In the following stage of the process, the questionnaires were divided into three parts as follows: the general information of respondents, longan growers' production, and the impacts of COVID-19 on post-harvesting and sales. For the secondary data, relevant documentations were reviewed, then appropriate data were used for systematic mapping, further data validation, and content analysis.

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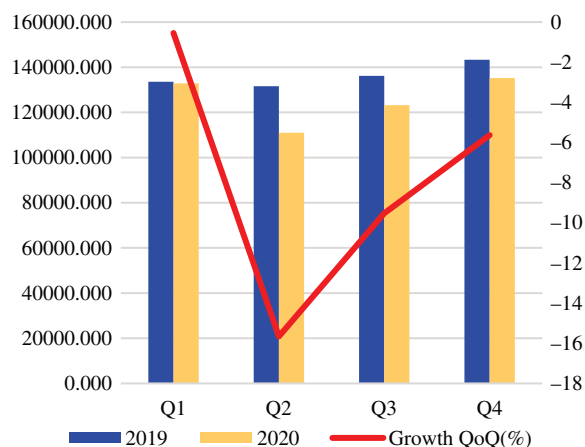


Figure 4: Nominal GDP and growth rate.

Source: CEIC Database (2020).

4. Results

4.1 Overview of macro-level analysis

The economic impact of COVID-19 on Thailand's economy has been devastating because of Thailand's openness to trade, food export, and tourism. As demonstrated onwards, Figure 4 presents the quarterly nominal gross domestic product (GDP) and GDP growth rate of Thailand. In the first quarter, the outbreak had trivial impact on the nominal GDP with a decrease in growth rate -0.55% , while in the second quarter of 2020, it dramatically decreased from USD 131.6 to USD 111 million or shrank by -15.65% compared with the same quarter in 2019, which is the sharpest decline in the Southeast Asia region. However, after Thailand controlled the spread of COVID-19 and implemented a gradual easing of the restrictions, the value of nominal GDP in the third and fourth quarters increased to USD 123 million and USD 135 million, respectively. However, this was the fourth consecutive quarter of contraction in the GDP, but the smallest decline since the Q1, as an improvement in both global demand and domestic activity. Considering the whole of 2020, the economy shrank by 7.78% , the sharpest drop since the 1998 Asian financial crisis.

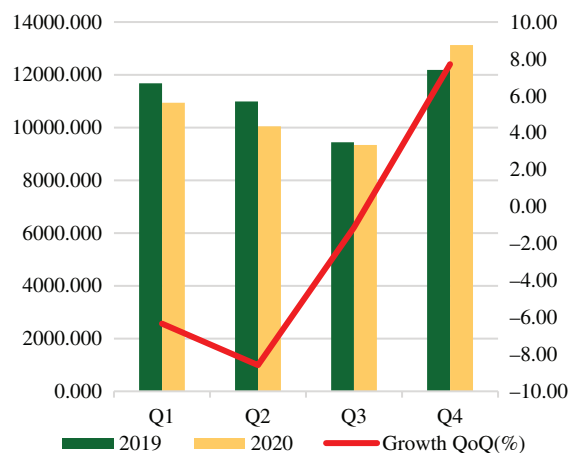
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Figure 5: Agriculture, forestry, and fishing nominal GDP and growth rate.

Source: CEIC Database (2020).

Figure 5 presents the GDP of Agriculture, Forestry, and Fishing (AFFGDP) of Thailand. In the first quarter, AFFGDP decreased from USD 11.7 million to USD 10.9 million, accounting for -6.33% compared with the same quarter in 2019. In the second quarter, the AFFGDP still declined from USD 10.9 million of 2019 to USD 10.0 million or -8.58% . However, the AFFGDP has recovered in the third quarter, restoring to USD 9.34 million, close to the value of AFFGDP in 2019 or an increase of 1.13% . Surprisingly, the AFFGDP in the fourth quarter increased at 7.72% to USD 13.13 million. According to the trade statistics from the Ministry of Commerce (2020), in 2020, the agricultural products with the highest export value are the preparations of meat with export valued USD 6.55 billion, followed by edible fruit and cereals, respectively. Remarkably, even during the COVID-19 pandemic, the edible fruit and cereals were higher than 2019 with an export value of USD 4.16 billion, an increase by 10.86% .

4.2 Thai fruits

The COVID-19 outbreak has had a strong impact both economically and socially on the stockholders of the different activities along the Thai fruit

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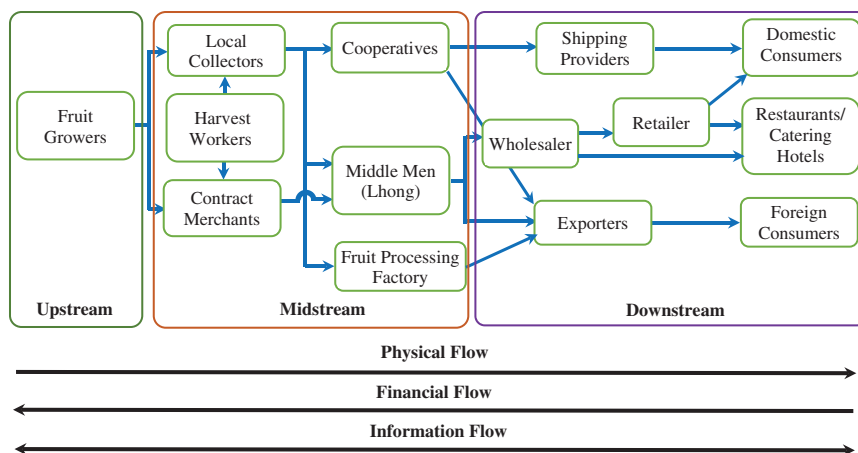


Figure 6: General Thai fruit supply chain.

supply chain. Figure 6 explains the stakeholders in the Thai fruit supply chain. In the upstream, the person who plays the critical role is the fruit grower or farmer who plants the fruit trees commercially. At the middle stream, there are many players undertaking different activities, namely, collectors, fruit harvest workers, contract merchants (who would buy in advance), middle men (Lhong), exporters and traders, agriculture cooperatives, wholesalers and retailers, domestic shipping providers, and supermarkets. The players downstream are consumers, restaurants, catering units, hotels, and foreign imports.

4.2.1 Fruit growers or farmers

Although many kinds of Thai tropical fruits and harvesting season began in March together with pandemic controllability of China, the epidemic without borders outside China has become more serious. China has continued to tighten measures to curb foreign travel in and out, and to encourage all airlines to reduce passenger flights to China. In the meanwhile, the government of Thailand announced a nationwide curfew starting on 3 April 2020, all borders had to be closed, and all normal commercial international flights in Thailand were suspended. This condition affected the arrival of fruit buyers, brokers, and traders from abroad, especially China

184 *Revitalising ASEAN Economies in a Post-COVID-19 World***Table 1:** On-season fruit price index 2019–2020.

Month	Durian			Rambutan			Mangosteen		
	2019	2020	% Change	2019	2020	% Change	2019	2020	% Change
May	539.79	439.51	–18.58	190.48	115.28	–39.48	259.22	138.69	–46.50
June	578.81	463.78	–19.87	218.79	172.42	–21.19	240.05	115.97	–51.69

Source: Office of Agricultural Economics (2020b).

and Vietnam who come to Thailand to evaluate both the quantity and quality of the product each year, and to negotiate and enter into trade contracts. In addition, the pandemic has changed international trade patterns and behaviours as more and more people took to digital media such as Line or WeChat. As a consequence, the restrictions resulted in fewer contracts for fruit purchases and contracts of lesser size and value. It is empirically demonstrated that the total demand for on-season fresh fruit and the farm-gate fruit prices had been reduced as shown in Table 1, the on-season (i.e. May and June) fruit price index of durians, rambutans, and mangosteens.

4.2.2 *Harvest workers*

Farmers and local workers indicated that they are not affected by the COVID-19 epidemic and curfew measures as they are able to maintain social distancing while harvesting, and their working time is during day-time. However, the affected groups are foreign workers and ethnic workers, particularly in the fruit harvesting season in which they have to work laboriously to meet harvest scheduling. The lockdown measures, setting up of the border control, prevented foreign workers (from Cambodia and Myanmar) from crossing the border at harvest time. From in-depth interviews with fruit farmers in Chiang Mai and Chanthaburi provinces, it was found that the effects of COVID-19 caused migrants to return to their home countries and they could not cross the border back to work in Thailand due to labour mobility prohibition across the border. Moreover, a key problem in relation to the hill tribe workers is that they cannot leave their hometown or village for job seeking during the fruit harvest season. This problem has raised the scarcity of labour supply in the short harvesting season. It has become a limitation for contract merchants, local collectors, and lhong, leading to wage increases.

4.2.3 *Wholesaler and retailer*

A variety of measures which the government applied in the early stages of the COVID-19 pandemic, such as the curfew pass and work from home arrangement, had a drawback on domestic fruit consumption, transportation, and logistics. After applying government work from home guidance, fruit sales in both the fresh market (wholesale and retail markets) and the modern trade market (supermarkets) fell. A number of businesses have temporarily ceased their operations, causing local workers to lose their jobs and move back to their hometowns corresponding to declining demand and sales of fruits. In addition, the recent COVID-19 crisis has led consumers to change their purchasing behaviour, substituting traditional markets for online shopping and delivery platforms such as Grab, Food Panda, and Line Man. Furthermore, the suspension of cross-provincial and international travel also has a tremendous impact on hotels, restaurants, and catering; significantly related to the declining demand for fruits from wholesalers and retailers when these businesses were temporarily shut down.

4.2.4 *Fruit processing business*

The COVID-19 outbreak has heightened the need for processed fruit (canned fruit or dried fruit) to continually maintain food security at the household level. The effects are multidimensional. For example, the sluggish global economy resulting from COVID-19 decreased the orders of processed pineapples or canned pineapples in Thailand from foreign markets, related to the reduction of production capacity. Moreover, the manufacturing labour shortage, which occurred with labour mobility restriction measures, leads to the delay in purchasing raw materials for farmers. Consequently, the fresh pineapple factory price was lessened.

4.2.5 *Transportation and logistics*

The government's stringent measures in the early stages of the COVID-19 pandemic, such as a strict night-time curfew, severely affected the transportation and logistics systems needed to transport fruits from orchards to fresh wholesale markets, local fresh markets, and foreign countries. The

acute problem of fruit exporters by land transport is the 14-day quarantine requirement for workers and truck drivers if they cannot return to Thailand within 5 hours, and some international border checkpoints required they remain for a period of 3 days and nights. The issue in relation to these situations arises from a shortage of containers. As a matter of fact, at some border checkpoints, the government officer requires COVID-19 health certificate for drivers and workers to cross international borders. In particular, at the Vietnam checkpoint on route R12 (connecting Thailand, Laos, Vietnam, and China), the officer randomly examines the medical certificate of the driver transporting durians without prior notice. The transport companies and exporters were unable to provide documents in a timely manner; therefore, fruit container vehicles could not cross the Cha Lo border, Vietnamese checkpoint. Similarly, the sea freight had to encounter insufficient containers for the products. In case of air freight, lower number of flights and higher air freight rates dominated the higher export and overseas retail prices.

4.3 Longan case study

Longan is a tropical fruit that is classified as the main economic fruit and product champion of Thailand that can be produced for domestic consumption and export, with the second largest export value after durian, holding China as the main export market. While durian export values increased compared with the previous year, longan export values dropped significantly. Figure 7 displays the value and month-over-month growth rate of fresh longan export. In 2020, the total longan export value of Thailand declined to USD 256 billion compared with USD 407 billion of 2019 or equivalent to -37%. The longan export value growth rate is negative every month except December, when a log of off-season longans is exported to China before Chinese New Year's Eve. The longan production in Thailand has been categorised into two types, namely in-season and off-season. Likewise, their supply chains from farms to consumers have been affected by COVID-19 shocks. Therefore, the research question is what are the economic and social impacts of COVID-19 on in-season longan supply chain in Thailand.

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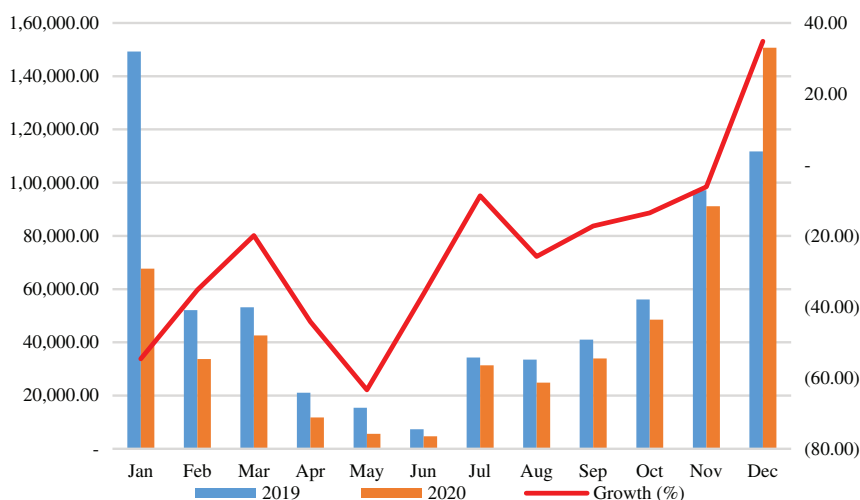


Figure 7: Value and growth of fresh longan export.

Source: Trade Map (2020).

Most of the in-season longan cultivation areas are in Northern Thailand, especially Chiang Mai and Lamphun provinces. In 2019, Thailand had 192,268.48 hectares of longan plantation area, rising by 1.46% from 2018. The longan output in 2019 was 1.01 million tons, declining from 2018 by 3.53%. As mentioned earlier, there are two types of longan production in Thailand: in-season longan and off-season longan. In-season longan refers to the longan production in natural conditions; the flowering commences in late December to late February, while the harvest time is from late June to late August.

Based on Tansuchat *et al.* (2016) and Panmanee *et al.* (2018), Figure 8 represents the on-season longan supply chain mapping. There are many stakeholders in the longan supply chain classified by activities as follows: (1) upstream: longan growers or farmers, (2) midstream: harvest workers, collectors, contract merchants, middlemen (Lhong), agriculture cooperatives, and longan processing factors, and (3) downstream: wholesalers and retailers, domestic shipping providers, and exporters. Each stakeholder has different roles, functions, and priorities.

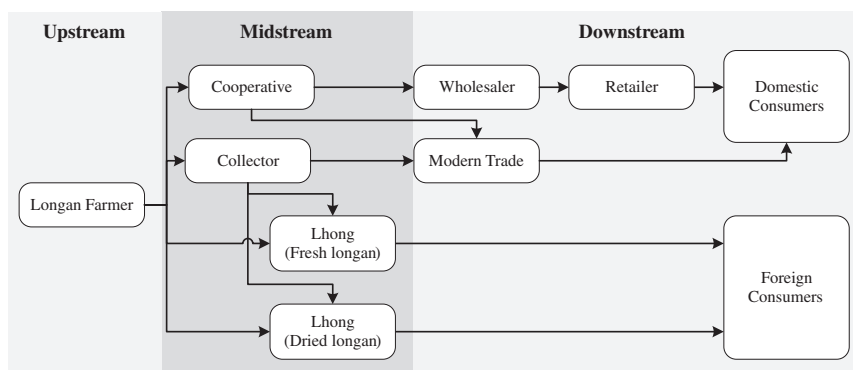


Figure 8: On-season longan supply chain mapping.

To clarify, the upstream starts with farmers growing longan as vigorously as possible by pruning together with providing water and farm management to protect longans from weeds, pests, and diseases. The coronavirus outbreak (COVID-19) and implementation of various government measures coincided with the on-season longan harvesting periods (June–August). Measurements were taken using 200 questionnaires collected from longan farmers in both Chiang Mai and Lamphun provinces. Of the study samples, the respondents are male (66.5%) and female (33.5%). The average age of respondents is 59 years old, and most of them are in the range 51–60 years old (33%). Most respondents (67%) have longan cultivation as their principal occupation. They have more than 15 years of experience in on-season longan farming. The average cultivation area is 3.97 rai or 0.64 hectares.

In response to the harvesting and selling of products, it was found that 56.28% of them have the capability to sell their products, comprising 39.7% of farmers fending for themselves and the rest of them undertaking the contract-merchant process. The majority of respondents (70.89%) harvest fresh longan by themselves, while the minority (29.11%) employ harvest workers. Complying with the COVID-19 pandemic situation and lockdown measure leading to the scarcity problem of the seasonal foreign workers, 95.65% of harvest workers are local workers. When the subjects were asked about the product sales under harvesting and sorting worker scarcity issue, the result indicated that they sold in the loosened longan

Table 2: Number of farmers affected by COVID-19 (province).

Province	Lamphun		Chiang Mai	
	Frequency	Percentage	Frequency	Percentage
Affected by COVID	89.00	89.00	99.00	99.00
Not affected by COVID	11.00	11.00	1.00	1.00
Total observation	100.00	100.00	100.00	100.00

Table 3: Effects of COVID-19.

Effects	Lamphun		Chiang Mai	
	Number	Percentage	Number	Percentage
Longan cheaper	85.00	95.51	98.00	98.99
No merchants	16.00	17.98	5.00	5.05
Hard to sell longan.	6.00	6.74	6.00	6.06
Other	5.00	5.62	7.00	7.07

Note: Respondents can choose more than one effect.

(detached from their inflorescences) 52.69%, longan in basket 37.63%, and bunched longan 9.68%. This is because the bunched longan and longan in basket required more skilled and meticulous labour to handle than loosened longan.

According to Table 2, 188 respondents or 94% reported that they experienced COVID-19 effect. The respondents agreeing on the decrease in longan price were 80.26% (Table 3). This corresponds to lower farm-gate prices of grade A fresh longan, compared with rates in the same month of 2019, as shown in Figure 9. In addition, 9.21% of respondents faced the problem of none of the collectors and merchants being able to buy fresh longan. Therefore, farmers decided to switch their selling pattern from longan in basket or bunched longan to a loosened longan sale. Normally, the Office of Commercial Affairs Lamphun (2020) reported that the farm-gate price of longan in basket is about 30–32B/kg. and bunched longan is about 26–28B/kg, which are higher than loosened longan (19–20B/kg). Consequently, farmers are more likely to suffer from deficiency compared to selling on-season longan.

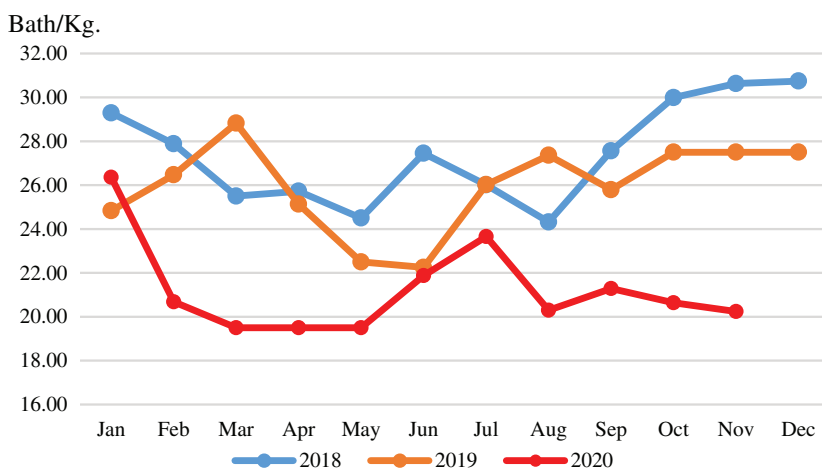
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Figure 9: The price of grade-A longan in bunch form between 2018–2020.

Source: Office of Agricultural Economics (2020b).

In the harvesting process, longan picking is the activity requiring skilled labourers, so a large number of skilled labourers are needed for collectors, contract merchants, or farmers. In Northern Thailand, most of them are foreign workers from Myanmar and hill tribe workers. Measures to prevent the spread of COVID-19 include controlling the labour mobility across the border, thus obstructing them to do the harvesting jobs. Furthermore, during the harvesting season, from July to August, or the rainy season, longan should not be picked as the additional risk of breakdown after harvest. In sum, labour shortage raised problems of late harvesting, resulting in crop damage and lowering farmers' income, leading to additional financial burden and household debt.

The problems of cross-provincial travel and the shortage of harvest workers affected the collector/contract merchant, causing the decline of workers and financial liquidity problems. Adversely affected domestic retailers and wholesalers as a result of curfew restrictions are unable to open because of the public concerns bringing about the change of consumer buying behaviour. Hence, some operators comply with selling fresh longan online by means of website or Facebook, and delivering them through domestic shipping providers, such as Thailand Post and Kerry in lieu of traditional marketing. Apparently, the online market is fairly

popular among consumers as it helps in avoidance of gatherings, which reduces the risk of contracting the virus.

The processed longan with the most economic value is dried longan (dried longan with shell, and gold dried longan without shell), and the major in-demand import country is China with 68.88%. From interviewing with dried-longan peeling workers, it was found that the situation of the COVID-19 pandemic resulted in low product demand. It is confirmed by the report of the agriculture office of the Royal Thai Consulate-General, Shanghai, that the COVID-19 pandemic and drastic measures to suppress the pandemic have reduced demand for dried longan, resulting in a large amount of dried longan remaining in Chinese stock. In addition, the dried longan for domestic consumption can be produced and supplied from three provinces, namely Guangdong, Fujian, and Guangxi, resulting in a decrease in demand for imported dried longan from Thailand. Moreover, the uncertainty of the pandemic situation heightened Chinese consumers' awareness of their spending being more focused on essential products for life purchase; in consequence, markedly declining demand for dried longan. Another problem that arose was concerned with the issue of trade agreements with the importers. In general, the importers or brokers originally commuted to Thailand to cope with the situation on fresh and dried longan, including negotiating the price, quality, and quantity of the dried longan. Despite the 14-day quarantine for all arrivals, trade negotiations of foreign traders, brokers, and importers were impracticable. Cross-border negotiations were altered into trading by means of online media, such as WeChat, Line, and Email. Unfortunately, exporters have to confront excessive pricing with excessively complicated procedures, such as creating animated videos to serve the purpose of marketing campaign and providing consumer product safety documentation for the decision concerning requirements. The export quantity of dried longan in the previous 9 months of 2020 thus decreased to 112.4 tons or equivalent to -1.8%.

Based on the impact of COVID-19 from the respondents, the Thai government has eased the measures to alleviate the affected longan farmers satisfactorily correlating with and further adopting these following measures. In the first place, the government generates the financial remedy measures for longan plantations compensating THB 2,000 per rai with the conditions of not more than 25 rai in order to comprehensively compensate other farmers. Second, launching the Online to Offline (O2O)

model project under the concept of modern commercial agriculture model aims to accelerate the distribution of fresh on-season longan in the northern region by approximately 30% of the total longan production. The O2O model project operates under the collaboration of private e-commerce companies and Thailand Post. Third, the Department of Cooperative Promotion would buy longan from members of the cooperative at a leading market price and distribute them for sale across provinces and across regions. Fourth, establishment of "Longan Institute" in the form of Agritech and Innovation Centre (AIC) and Centre of Excellence (COE) would conduct research and development aiming to reduce costs, focus on processing, add value to longan, and reduce dependence on exports as a sustainable solution.

4.4 Discussion

The results of the study confirmed that the COVID-19 pandemic and government measures had adversely impacted the agricultural and food supply chain system, which comprises many activities included in a "farm to fork" such as farming, harvesting, process management, packaging, transportation, distribution and marketing, and demand management, in at least three ways. Firstly, it is the demand channel. The COVID-19 pandemic destroys consumer confidence and creates uncertainty conditions in the country's economy. Therefore, people will stay connected to their homes and work from home to reduce the chance of contracting the disease. The consumption expenditure then declined, and shopping behaviour changed to online platforms. The reduction in consumption expenditure affects gross demand. For instance, international tourism is banned, affecting the tourism industry and tourism-related industries such as food and beverages industry, aviation industry, hotel and accommodation industry, and retail industry. Secondly, supply channels are changed. To demonstrate, gross demand reduction directly affects real sector production, employment, a considerable drop in sales, and business operations. Ceasing business and reducing employment has a direct impact on the income and living condition of household members. Lastly, the concern of supply chain channels is growing. The nation-wide lockdowns and safety measures from COVID-19 created the agriculture supply chain risks due to

harvest labour shortages for fruit growers, agriculture supply chain disruption, shortage and fluctuation in the supply markets, inadequacy of the logistics providers, restrictions on interprovincial and interstate transportation, and rising of logistics cost.

In case of longan, which is a tropical tree species that produces edible fruit, the COVID-19 pandemic destroyed many activities and stockholders. Nevertheless, the analysis did not confirm any significant differences between the overall fruit export. The result pointed to the strategy that farmers altered their products in terms of the loosened longan instead of bunched longan and longan in basket. It also created much risk in supply chain such as supply-side risks due to harvest labour shortages for fruit growers, fresh longan over supply, and reduced farm-gate price; demand-side risk due to unanticipated consumer demand, reduced purchasing power of domestic consumers, changes in food safety requirements, logistics; and transportation risk due to underperformance of the logistics providers because of nation-wide lockdowns and safety measures.

5. Conclusion

This chapter has documented and analysed the impacts of COVID-19 pandemic on Thai fruit sectors along with studying a special case in longan with its dominance. The findings of this study indicate that the variety of COVID-19 measures have adversely ripple effects on the stakeholders of Thai fruit supply chain. The lockdown and containment measures dwindle the fruit trading volume, on-season fruit export prices, farm-gate prices, leading to the increase in skilled harvest labour shortage, transportation and logistics cost, and higher food safety measures and requirements. The upshot of the current farmers' income and future production planning is under consideration. This possibility also obliges every stakeholder involved in the fruit supply chain sectors. In the case of longan, coronavirus and preventive measures point to the considerable impact towards the idea of business operations and exports of longan. This research also underlined the economic impact in terms of the lower demand for fresh on-season longan, bringing about the lessened prices, lower farmer's income, and reduced financial liquidity of longan-related businesses, especially farmers and merchants. Moreover, it is crucial to

note that the problems of supply-chain disruptions consist of logistics delays and jams, customs clearance delays, logistics and transportation costs, and changes in product distribution channels.

The results of the study can lead to policy recommendations. First, it would be necessary for the government to provide financial capacity with low interest rates for the next season's production. Second, promoting learning and training (reskilling, upskilling, and new skilling) courses about careful advance planning, fruit supply chain management, food safety, and traceability system are crucial for farmers and stakeholders with emphasis on digital skills; for instance, promoting Climate-Smart Agriculture (CSA) and Good Agricultural Practices (GAP); building high-value food industries such as in processing, packaging and retailing; and enhancing Food Value Chain (FVC) approach to food safety, security, and quality by promoting Public and Private Partnership (PPP) based on FVC and CSA. Third, food safety will become a mega trend in the future. Entrepreneurs in the agriculture and food supply chain should pay more attention and the government should accelerate the search for ways to promote safe food production, such as raising awareness of the importance of safe food for consumers. If consumers are willing to pay more, it will increase incentives for farmers to produce safer agricultural products. Fourth, the public and private sectors should try to optimise the customer's understanding about the correspondence between COVID-19 and fruit and vegetable products to reduce consumption anxieties. Fifth, although the restrictions of cross-border labour should be applied under the pandemic situation, liberalisation with the regulation of internal labour migration, especially fruit picking workers across provinces, should be taken into consideration. Lastly, the government should promote convenient export and import of agricultural and food products, and enhance regional and international cooperation to alleviate trade restrictions on agricultural and food products from measures to prevent the spread of COVID-19.

Acknowledgement

This research is a part of the study performed within the framework of research title "Food Loss Assessment of Longan Products for Identifying

the Food Loss Reduction Measures along the Whole Value Chain and Reporting SDG 12.3.1" which was made possible by a grant from the Agricultural Research Development Agency (Public Organisation) under Ministry of Agriculture and Cooperatives.

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